Cochran Boiler

Cochran boiler is efficient than the simple vertical boiler and it was founded by Cochran & Co. of Annan. In this boiler, the fire tubes are placed horizontally and the hot gases are surrounded around them for the generation of steam.

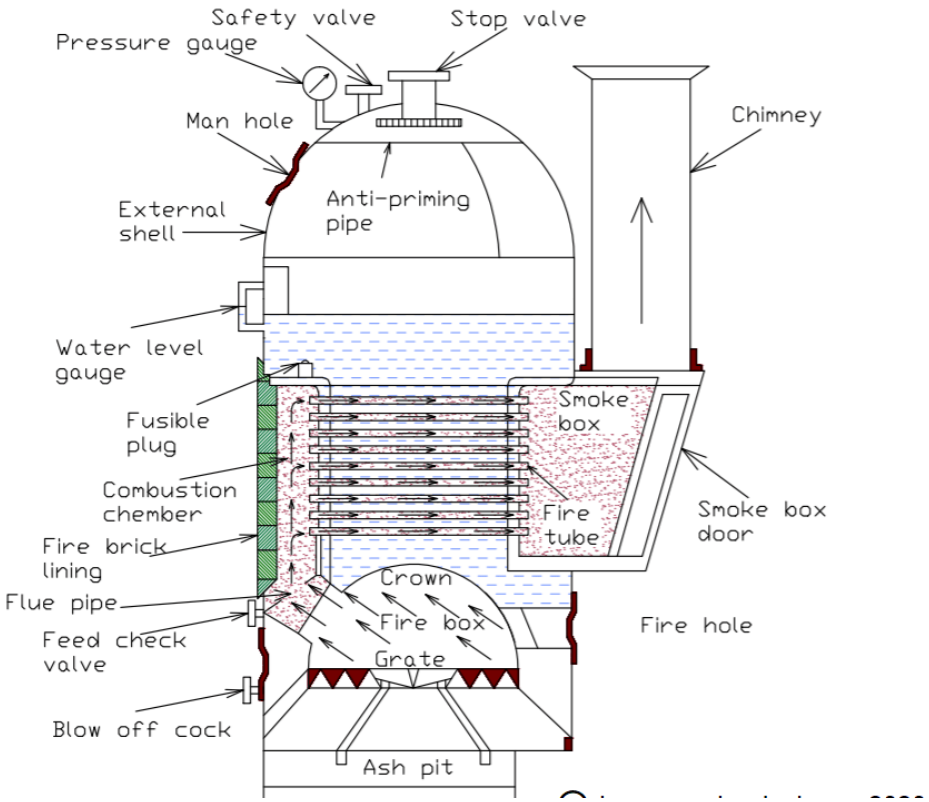
In this article, we will discuss **Cochran Boiler: Definition, Parts, Working, Specification, Applications, Advantages & Disadvantages** in a detailed way.

Cochran Boiler Definition:

Cochran Boiler is one of the best types of vertical, multi-tubular boiler which has a number of horizontal fire tubes used for the generation of steam.

Line Diagram of Cochran Boiler:

The diagram of Cochran Boiler is shown below.



cochran boiler

Parts of Cochran Boiler:

All the parts of Cochran Boiler are explained below:

1. Shell
2. Combustion Chamber
3. Grate
4. Ash Pit
5. Fire hole
6. Furnace
7. Flue Pipe
8. Fire brick Lining
9. Fire Tubes
10. Chimney
11. Manhole
12. Smokebox
13. Hemispherical Dome

An Explanation for the parts of Cochran Boiler:

The detailed explanation for the parts of Cochran Boiler is as follows.

1.Shell:

It has a cylindrical drum with a hemispherical dome at its top.

2.Combustion Chamber

Burning of fuel takes place in the combustion chamber.

3.Grate

It is the sectioned platform on which the fuel(Coal) is burnt.

4.Ash Pit:

Ash is collected from the burnt fuel placed on the grate which will deposit into the Ash pit.

5.Fire hole:

It is the hole provided to the shell to fire(burn) the fuel.

6.Furnace:

It is the region where combustion takes place and without a furnace, it is not possible to get the steam.

7.Flue Pipe:

The hot gases enter into the combustion chamber through the flue pipe only. It is a small passage that connects the combustion chamber and firebox.

8.Fire Brick Lining:

This helps in the combustion of the fuel. After striking brick lining, the flue gases pass through the horizontal tubes.

9.Fire Tubes:

This boiler has multi-tubular fire tubes. The fire tubes help in the exchange of heat from the hot flue gases to the water surrounded around it. The hot flue gases from the combustion chamber travel to the smokebox through these fire tubes only.

10.Chimney:

The chimney is attached to the smokebox which is used to transfer the flue gases coming from the fire tubes, to the environment.

11.Manhole:

A manhole is provided for inspection and cleaning of the boiler.

12.Smokebox:

It is used to collect the flue gases and unburned fuel coming out from the combustion chamber.

13.Hemispherical Dome:

The steam is deposited in this dome after its generation from water.

Other Boiler mountings and accessories attached to this boiler are:

Pressure Gauge:

It measures the pressure of steam inside the boiler.

Water level Indicator:

The water level in the boiler is indicated by the water level indicator.

Safety Valve:

It blows off the steam when it reaches beyond the safety level inside the boiler.

Blow off Cock:

It is used to blow off the impurities present in the boiler water.

Working Principle of Cochran Boiler:

The fuel(coal) is to be placed on the grate and is to be burnt from the fire hole. The air enters into the combustion chamber from the atmosphere.

The fuel is to be burned in the Firebox and the smoke, hot flue gases emerges into the combustion chamber through the flue pipes.

From the combustion chamber, hot gases enter into the fire tubes. As the water is surrounded by the fire tubes, exchange of heat takes place between them.

Due to the hot gases, the water converts into steam and moves upward to the hemispherical dome.

The Anti-priming pipe absorbs the water molecules from the steam collected in the hemispherical dome and makes it as dry steam which can be sent to the turbines to do work through the steam stop valve mounted on the top of the hemisphere.

The burnt gases and smoke is discharged to the atmosphere through the chimney.

The burnt fuel is to be collected into the Ash pit and the impurities which are present in the water will be discharged out through the Blow off cock.

A fusible plug is also provided at top of the combustion chamber. When the temperature of the combustion chamber crosses the permissible level, the fusible plug melts and the water enters into the furnace of the boiler and stop the fire. In this way, we can avoid accidents in the industry.

In this way, the steam is used to drive the turbines.

Specifications of Cochran Boiler:

The dimensions, working pressure, heating surface, capacity and efficiency of the Cochran Boiler are as follows.

- Shell diameter:b2.75m
- Height:5.79m
- Steam Capacity: 3500 kg/hr (maximum capacity
- Tubes diameter: 6cm
- Working Pressure: 6.5 bar (max pressure 15 bar)
- Heating surface: 120 m2
- Efficiency: 70 to 75%

Applications of Cochran Boiler:

- It is used in refining units
- It is used in paper and pulp manufacturing plants.
- It is also useful in chemical processing plants.

Advantages of Cochran Boiler:

- Low initial installation cost.
- Transportation of the Cochran boiler is easy.
- Easy to operate and handle.
- It requires less floor area.

Disadvantages of Cochran Boiler:

- Low rate of steam generation.
- It has a limited pressure range.
- The installation requires a large area due to its vertical design.
- Inspection and maintenance are difficult.

Lancashire Boiler is a horizontal type and stationary fire tube boiler. This boiler was invented in the year 1844, by William Fairbairn.

This is an internally fired boiler because the furnace uses to present inside the boiler. This boiler generates low-pressure steam and it is a natural circulation boiler.

It has high thermal efficiency of about 80 to 90 percent.

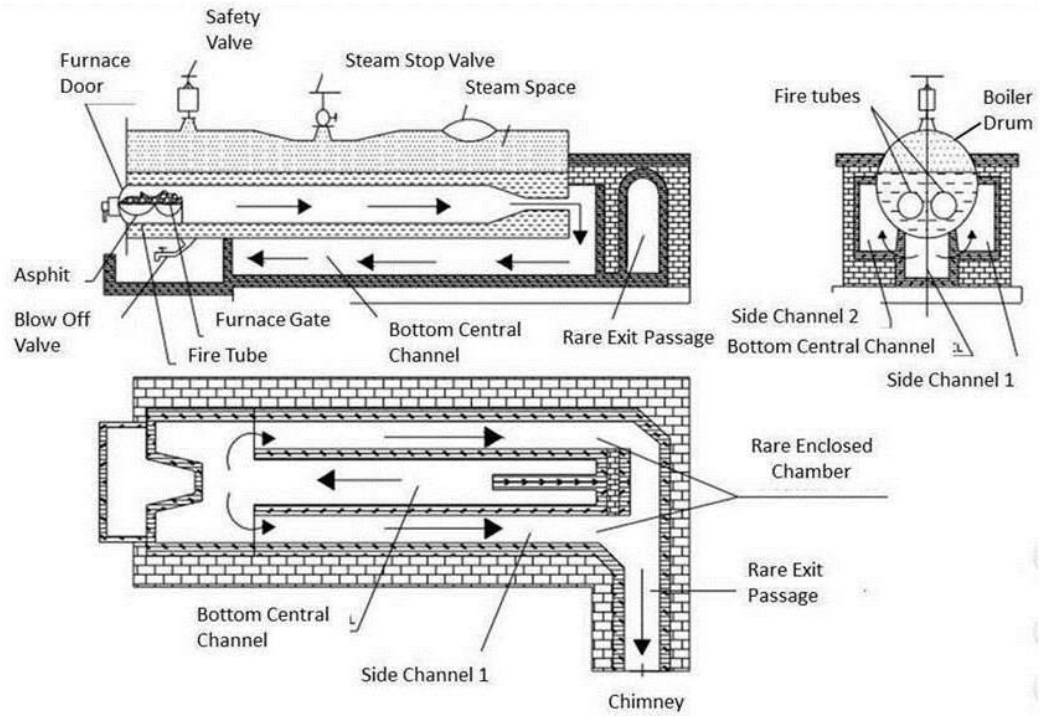
The size is approximately 7-9 meters in length and 2-3 meters in diameter.

It is mostly used in locomotive engines and marines etc.

Parts of a Lancashire Boiler:

A Lancashire Boiler consists of following parts:

- *Safety valve*
- *Pressure Gauge*
- *Feed check valve*
- *Water Level Indicator*
- *Blow off valve*
- *Steam stop valve*
- *Manhole*
- *Fire door*
- *Fusible Plug*
- *Ash pit*
- *Economizer*
- *Air preheater*
- *Superheater*



Lancashire Boiler

Lancashire Boiler

Safety valve:
The safety valve is used to blow off the steam when the pressure of the steam inside the boiler exceeds the working pressure.

Pressure gauge:
The function of the pressure gauge is to indicate the pressure of the steam inside the boiler.

Feed check valve:
It stops and allows the flow of water inside the boiler.

Water level indicator:
It indicates the level of water in the boiler. It is placed in front of the boiler. Two water level indicators are used in the boiler.

Blow off Valve:
Its function is to remove the sediments or mud periodically that is collected at the bottom of the boiler.

Steam stop valve:
The function of a steam stop valve is to stop and allows the flow of steam from the boiler to the steam pipe.

Manhole:
The hole is provided on the boiler so that a man can easily enter inside the boiler for the cleaning and repairing purpose.

Fire door:
This is used to ignite the present fuel inside or outside the boiler.

Fusible plug:
It is used to extinguish the fire inside the boiler when the water level inside the boiler falls to an unsafe level and prevent an explosion.

It also prevents the damage that may happen due to the explosion.

Ash pit:
The Ash-pit used to collect the ash of the burnt fuel.

Working Principle of Lancashire Boiler:
This Lancashire boiler works on the basic principle of the heat exchanger.

It is basically a shell and tube type heat exchanger in which the flue gases flow through the tubes and the water flows through a shell.

The heat is transfer from flue gases to the water through convection.

It is a natural circulation boiler that uses the natural current to flow the water inside the boiler.

Applications of Lancashire Boiler:
Lancashire Boiler can be used in several filed like:

- *The Lancashire boiler is used to drive steam turbines, locomotives, marines, etc.*
- *It is used in industries like paper industries, textile industries, sugar industries, tire industries, etc.*

Advantages of Lancashire Boiler:

These are some advantages of Lancashire boiler:

- *This has high thermal efficiency. Thermal efficiency is about 80 to 90%.*
- *This is easy to operate.*
- *It can easily meet the load requirement.*
- *Easy to maintain.*
- *Low consumption of electricity due to natural circulation.*

Disadvantages of Lancashire boiler:

Although a Lancashire boiler has some disadvantages and those are:

- *This is a low-pressure type boiler, so high-pressure steam is not produced.*
- *It has a limited grate area due to the small diameter of the flue tubes.*
- *The steam production rate is low. It is about 9000 kg/hr.*
- *Corrosion occurs in the water legs.*
- *This boiler requires more floor space.*